Mele Health

Mele Blockchain

The next financial system evolution

Times are changing fast. Technology has rolled over industries that resisted change: media, enterprise software, cable television, and most recently: stock brokerages. However, healthcare information management has been reluctant to change.

Patients are now demanding the same level of convenience, access, and flexibility they are used to as consumers of other products. Similarly, health providers need to free data from information silos in order to provide better service to patients.

Mele Health blockchain services offers bank-level security and convenience to the healthcare industry.

Existing Healthcare Infrastructure

01

Slow

Locating and sharing data

02

Expensive

Data management vendors

03

Friction

High levels of bureaucracy

04

Silo-ed

Data sits in various unintegrated

05

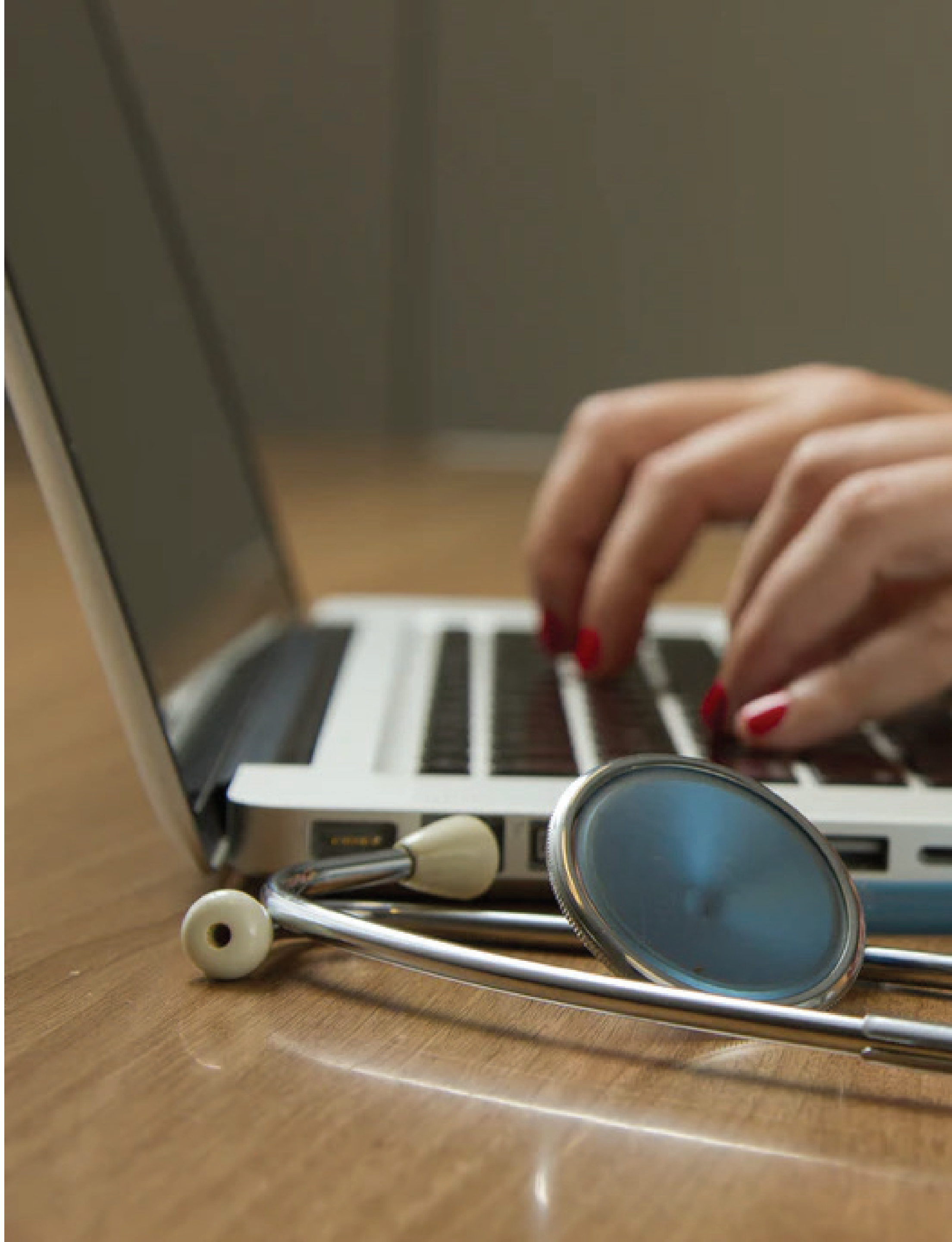
Inflexible

Lack of common protocols between data silos

06

Unreliable

No universally accepted audit trail

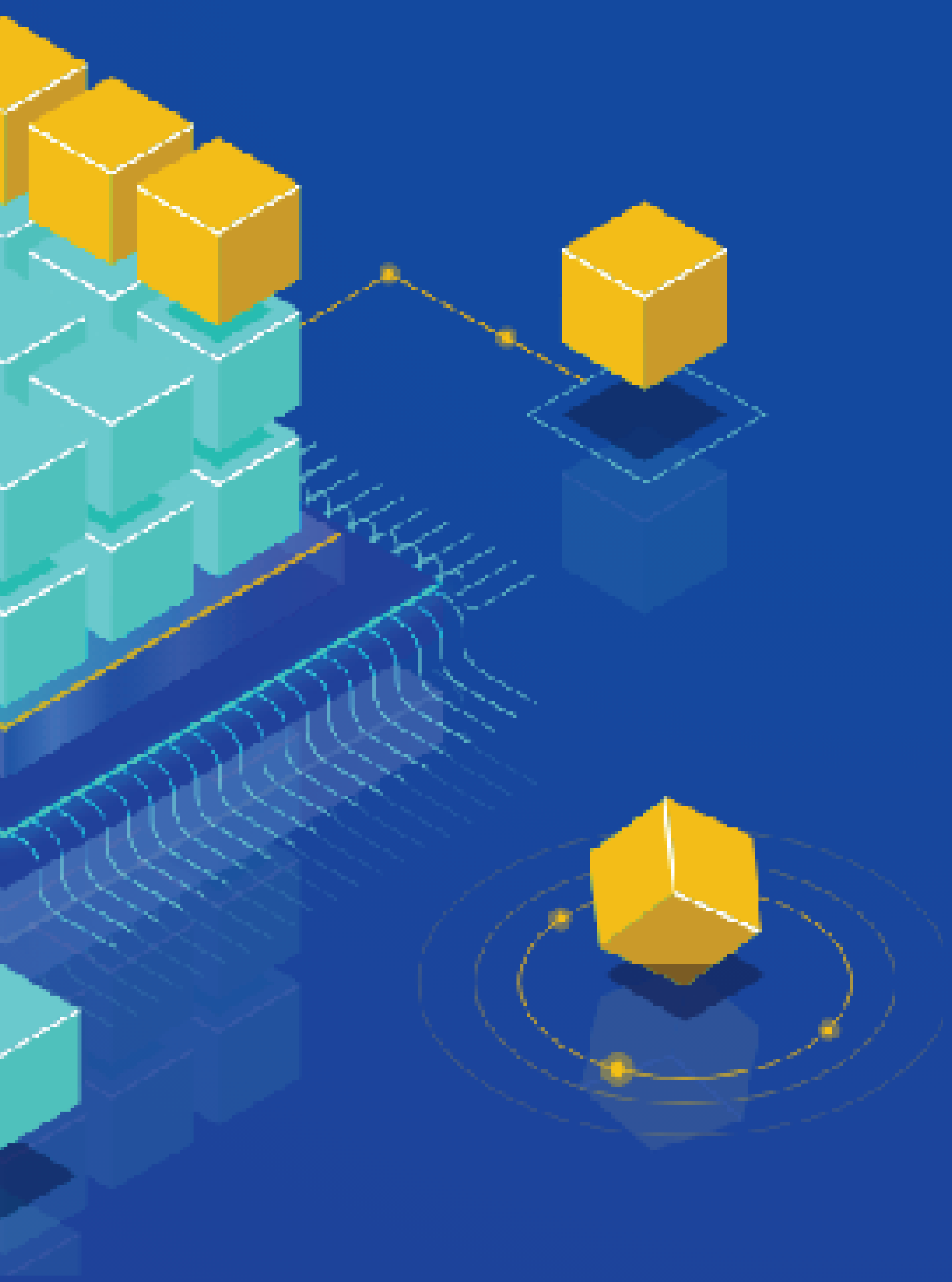


VISION رؤيـــة 2030
المملكة العربية السعودية
KINGDOM OF SAUDI ARABIA

Saudi 2030

Healthcare Modernization Objective for the Ministry of Health

- Enterprises to encourage privates sector investment in healthcare, municipal services, housing, finance, energy and education;
- Increase the efficient utilisation of available resources;
- Improve the efficiency and effectiveness of the healthcare sector through the use of information technology and digital transformation;
- Improve integration and continuity in service provision
- Establishment of a Unified Electronic System: >70 percent of Saudi citizens to have unified digital records



Blockchain Advantages

01
Accessible
Easy to locate and share data without silos

02
Decentralized
Data lives in multiple locations

03
Economic
One protocol, highly scale-able

04
Flexible
Unlimited levels of access and permis-sions

05
Transparent
Full provenance. Audit record of data access and updates.

06
Secure
Data protected by pre-agreed rules combined with the most advanced encryption technology

Use Cases

Electronic Medical Records
Locating and sharing data

Medical Billing
Data sits in various unintegrated silos

Healthcare/Vaccination Passport
data management vendors

Medical Billing
Data sits in various unintegrated silos

Health Data Exchange
Lack of common protocols between data silos

Clinical Trials
No universally accepted audit trail

HYPERLEDGER
INDY

Mele Health Blockchain Architecture

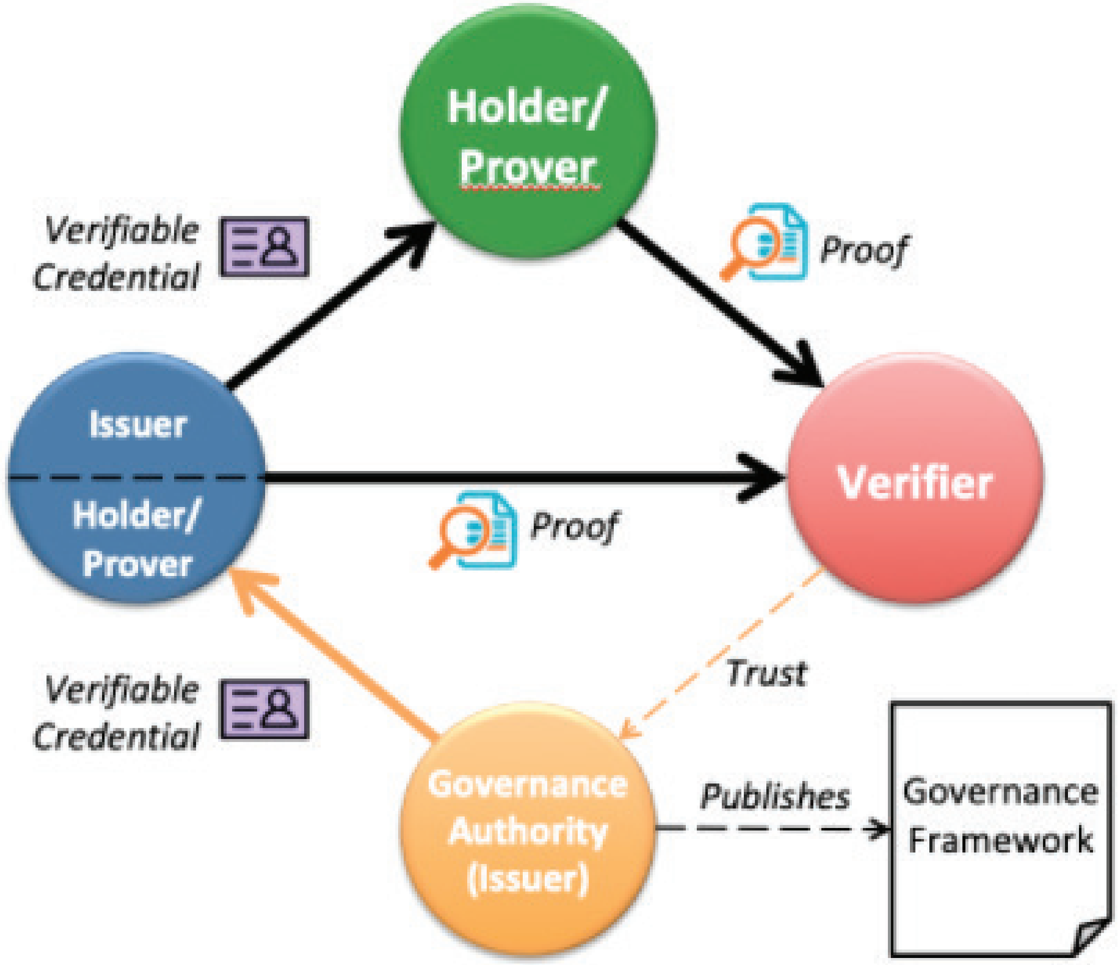
Built on the Hyperledger Indy Fabric

- Patient Control
- purpose-built for decentralized identity
- DIDs (Decentralized Identifiers) requiring no centralized resolution authority
- Verifiable Credentials in an interoperable format for exchange of digital identity
- Zero Knowledge Proofs: validation using small pieces of information without personal identifiers

Proposed Pilot: SSI Medical Digital Identity

Based on scheme implemented by our team for Messly in cooperation with the UK NHS

- Messley: UK temporary staffing agency placing doctors at NHS hospitals
- Stakeholders: Issuer, Holder, Verifier
- Trust Triangle: Verifiers must trust the Issuer:
- Decentralized Identifiers (DID): cryptographically identifiable identity for digital agents
- Governance framework: the set of business, legal, and technical rules for how SSI infrastructure is used. Set by Public Health Authority



How SSI Works

- Verifiable credentials
- Issuers, holders, and verifiers: the “trust triangle”
- Digital wallets: digital equivalent of our physical wallets
- Digital agents: apps or software modules that interact
- Decentralized identifiers (DIDs)
- Governance frameworks: set of business, legal, and technical rules for how SSI infrastructure is used

Electronic Medical Records (EMR)

Breaking down healthcare data silos

Electronic medical records were not originally designed to be used across institutions and over a lifetime. Patients interface with multiple institutions throughout their lives leaving the data that makes up their aggregate record to exist within numerous silos that do not efficiently interact with one another. The result is that historic data is lost, incomplete, or inaccessible. Blockchain technology makes possible a new standard architecture that breaks down the silos and enables more secure, flexible, and collaborative healthcare solutions.

Electronic Medical Records (EMR)

- Autonomy: patient controls the full record rather than the healthcare institution(s)
- Transparency: full, permanent record of interactions with the data
- Privacy: patient could maintain anonymity while interacting with medical institutions
- Interoperability: multi-party collaboration/interaction
- Authentication: Self Sovereign Identity (SSI) means participants control unique identifiers to create “Trust Triangle” for sensitive data and credentials
- Availability: records are easily accessible

Other use cases...

Credentialing

- Autonomy: healthcare professional controls their full record rather than the healthcare institution(s)
- Transparency: full, permanent record of interactions with the data
- Fast: instant credential sharing between jurisdictions as opposed to the time-consuming process now

Clinical Trials

- Privacy: patient could maintain anonymity while participating in large studies
- Accessible: data is no longer silo-ed but can be shared with approval of the patient
- Scalability: enables large datasets and the ability to query specific users for more access
- AI: machine-learning models can be trained on anonymized data
- Authentication: Self Sovereign Identity (SSI) means participants control unique identifiers to create "Trust Triangle" for sensitive data and credentials
- Availability: records are easily accessible

Team

Deep Blockchain Experience

Blockchain Developer

Blockchain research and development with a focus on Tendermint, Solidity, Ethereum, Cosmos SDK, and Hyperledger. Developed software solutions for microprocessors, core banking, the web, and money-handling equipment.

Product Manager

Led software projects using Agile software development in a variety of roles including Project Manager, Scrum Master, and Scrum Coach.

Developed enterprise apps, websites, microsites, and workflow systems for some of the largest companies in

Commercial Coordinator

Led token economics modelling for venture-backed services marketplace. Led financial budgeting, strategy, and analysis for US seed and venture funded companies in marketplace, blockchain, and b2b and b2c SaaS categories.

Front-end Engineer

Adaptive Front-End Developer with 11+ years of experience delivering powerful web and mobile solutions with a focus on blockchain tech. Skilled in development, design, and testing of multiple web-based applications incorporating a range of technologies.